

DENIS SAMATOV

Machine Learning Engineer & Technical Lead · LLM/RAG & Evaluation · Medical Imaging · Scientific ML

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SUMMARY

Machine learning engineer with 5+ years of applied ML across LLM/RAG systems, medical image segmentation and scientific computing. Technical lead for LLM/RAG systems in use (teams of 3–5 engineers): hybrid retrieval, source-grounded generation, MCP tool integrations and release-blocking evaluation gates. Built EPIFAT, state-registered cardiac CT segmentation software (Attention U-Net, Dice 0.91 on a patient-level split), and open-source research software for tensor-based reduced-order modelling and Bayesian inference. M.Sc. in Applied Mathematics and Computer Science.

PROFESSIONAL EXPERIENCE

Analytics and Machine Learning Department, MSUU

Machine Learning Engineer / Technical Lead | Apr 2024 – Present

- Lead architecture and delivery of LLM/RAG systems with teams of 3–5 engineers; three are in use: a course-catalogue assistant (CourseLLM), a housing knowledge-base assistant and an internal Telegram knowledge hub (pilot).
- Built hybrid retrieval (BM25 + Qdrant dense + title n-grams) with reranking and source citations over a 627-document / 1,462-chunk index; a 37-query protected retrieval regression reaches Hit@5 1.00, MRR@10 0.89 and nDCG@10 0.90.
- Made evaluation release-blocking: a 50-case RAGAS acceptance gate and LLM-as-judge checks in CI. The gate stopped a candidate release (answer relevancy 0.72 vs a 0.90 threshold) and drove fixes to exact-FAQ retrieval.
- CourseLLM: FastAPI service with streaming and queued endpoints, Redis-backed sessions and a concurrency semaphore; 18-case golden-QA release gate; load-tested at its configured limit of 7 concurrent requests (p95 4.6 s, overflow shed with HTTP 429).
- Lead LLM Adviser (in development, team of 5), a knowledge platform for geological and reservoir-simulation literature; delivered the ingestion layer: PaddleOCR-VL, DOC/DJVU/CHM-to-PDF conversion with provenance, immutable manifests, a PostgreSQL registry and Redis/RQ workers.
- Built MCP integrations giving agents governed access to Yandex Disk and Wiki (open source: 50+ typed tools, read-only by default, permission gates, audit logging); route between API and self-hosted LLMs (GigaChat, YandexGPT, DeepSeek, Gemma via Ollama).
- Set team engineering standards: design reviews, typed Python (mypy, ruff), automated tests and CI quality gates.

Heriot-Watt TPU Center

Data Scientist / Research ML Engineer (part-time) | Sep 2024 – Present

- Develop TBMD, an open-source PyTorch library for Tucker/HOSVD modal decomposition, QR-based sparse sensor placement and compressive field reconstruction (CI on Python 3.10–3.12, Zenodo DOI).
- Audited the arXiv v1 Brugge results as non-reproducible and rebuilt the study as a nested leave-one-scenario-out benchmark over 10 scenarios: partial factor sharing cut anomaly-relative error from 1.17 to 0.35 (pressure) and 0.84 to 0.47 (saturation) at 30 grid channels, while POD remained strongest at existing wells.
- Develop NAB (nab-bayes on PyPI), a Python toolkit for Bayesian parameter-space appraisal (MCMC / Gibbs sampling over Voronoi cells) with CI, a Docker image and 540 tests.

Cardiology Research Institute

Machine Learning Engineer | Mar 2021 – May 2026

- Built EPIFAT, state-registered software (Certificate No. 2025610317) for epicardial adipose tissue analysis on non-contrast cardiac CT: AI pericardium segmentation, HU-based fat extraction, manual contour and brush correction, 3D view and PyRadiomics features; desktop GUI and CLI with 470 tests.
- Trained U-Net-family models on a 62-patient / 5,276-slice dataset with a patient-level split and corrected metric aggregation: Attention U-Net reached Dice 0.91 (IoU 0.88) vs 0.90 for U-Net (single run).
- Co-authored peer-reviewed cardiac MRI radiomics studies (2024–2026), e.g. detection of post-infarction myocardium with Lasso logistic regression (AUC 0.76, 33 patients).

Earlier (2023): JINR START programme — ML for beam-parameter restoration at the NICA accelerator complex (Jun–Aug); TUSUR LPMITF — Data Scientist (Jun–Dec); Fintech Association — ML Engineer (Aug–Nov).

OPEN SOURCE

- Merged upstream: Weaviate Python client #2153 (generative DigitalOcean integration), DragonflyDB #8199 (RedisVL compatibility), IBM docling-pipelines #31 (side-effect-free dry runs), W&B rai-toolkit #24 (HR industry preset).
- Public repositories: tensor-based-modal-decomposition-method, yandex-workspace-mcp, llm-adviser-case-study (github.com/denis-samatov); nab-bayes on PyPI. EPIFAT and CourseLLM code is private; walkthroughs available on request.

SKILLS

LLM & retrieval: RAG, hybrid retrieval (BM25 + dense), reranking, GraphRAG, LangChain, LangGraph, MCP, tool calling, RAGAS, LLM-as-judge, Qdrant, pgvector, Ollama; API and self-hosted LLMs (GigaChat, YandexGPT, DeepSeek, Qwen, Gemma)

ML & computer vision: PyTorch, scikit-learn, U-Net / Attention U-Net, semantic segmentation, DICOM, PyRadiomics, OCR (PaddleOCR, Docling, Tesseract)

Scientific computing: tensor decomposition (Tucker/HOSVD), sparse sensor placement, compressive sensing, reduced-order modelling, Bayesian inference, MCMC

Engineering: Python (typed), FastAPI, PostgreSQL, Redis/RQ, Docker, GitHub Actions, pytest, mypy, ruff, MLflow, Weights & Biases, Prometheus

EDUCATION

Tomsk Polytechnic University — M.Sc., Applied Mathematics and Computer Science | 2024 – 2026

GPA 5.0/5.0. Thesis: tensor-based modal decomposition for reconstructing reservoir properties from sparse measurements.

Tomsk Polytechnic University — B.Sc., Applied Mathematics and Computer Science | 2020 – 2024

GPA 5.0/5.0. Thesis: automatic segmentation and radiomic analysis of epicardial adipose tissue in cardiac CT.

Professional development

TPU professional retraining diploma, Data Science and ML (2023–2024); Skoltech courses: LLM-Based Agents (2025), Generative Adversarial Models (2024).

PUBLICATIONS, IP AND RECOGNITION

- Samatov D., Merzlikin B., Shishaev G. Tensor-Based Modal Decomposition and Sparse Sensor Placement for the Brugge Field Simulation Model. arXiv:2607.09687, 2026 (revised manuscript in preparation).
- Non-contrast cine cardiac MR radiomics for the prediction of left-ventricular adverse remodeling in ischemic cardiomyopathy: a pilot study. Siberian Journal of Clinical and Experimental Medicine, 2026.
- Radiomics and machine learning to detect post-infarction cardiosclerosis on non-contrast cardiac MRI. Russian Journal of Cardiology, 2026.
- Radiomics analysis of cine-mode cardiac MRI to detect post-infarction left-ventricular lesions. Digital Diagnostics, 2024.
- EPIFAT — Certificate of State Registration of Computer Program No. 2025610317, 2025.
- Accepted oral, AI4X Accelerate, Singapore, 2026 · Winner, AI/ML track, FINODays hackathon · Special nomination winner, National Technology Olympiad (computer vision track).